The FB $\leftrightarrow$ GCIQL bridge

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1 The two agents, and the design decisions they expose

Both agents parameterize a goal/task value as the *same* normalized inner product and steer it with the *same* flow (FQL) actor. They differ only in a handful of design decisions. GCIQL learns a goal-conditioned optimal value by IQL:

$$Q(s,a,g)=\tfrac{1}{\sqrt{d}}\phi(s,a)^\top\psi(g),\quad \mathcal{L}_V=\mathbb{E}\big[L_2^\tau(\bar{Q}-V)\big],\quad \mathcal{L}_Q=\mathbb{E}\big[(r+\gamma(1-d)V'-Q)^2\big],\quad r=\mathbb{1}[s\approx g]-1,$$

where the expectile L_2^τ ($\tau>\frac{1}{2}$) is an implicit $\max_a$ (policy improvement). FB instead fits the discounted successor measure of a policy *family* $\{\pi_z\}$ — *no reward, no max*:

$$M^{\pi_z}(s,a,\mathrm{d}s')\approx F(s,a,z)^\top B(s')\rho(\mathrm{d}s'),$$

$$\mathcal{L}_{\mathrm{FB}}=\underbrace{\mathbb{E}[(F^\top B'-\gamma\,\mathrm{sg}[\bar{F}^\top\bar{B}'])^2]}_{\text{TD over random } s'\sim\rho}-2\,\mathbb{E}[F^\top B^+]+\lambda\|\mathbb{E}[BB^\top]-I\|_F^2,$$

with a post-hoc linear task readout $z_r=\mathbb{E}_{s'}[r(s')B(s')]$, $Q_z=F(s,a,z)^\top z$. The eight decisions that separate them (each a knob in the JAX GCIQL agent `gciql_flow.py`, or the FB torch agent `agents/fb/agent.py` for the FB-side rows) are:

	Design decision	GCIQL setting	FB setting	Knob
D1	Value objective	goal-reward IQL value-TD (expectile max)	successor-measure density (no max)	`measure_only`
D2	Critic form	bilinear $\phi^\top\psi/\sqrt{d}$ (or MLP)	bilinear $F^\top B/\sqrt{d}$	`value_arch`, `fwd_two_tower`
D3	Fwd-map task-cond	$\phi(s,a)$, goal is a net input	$F(s,a,z)$, z post-hoc	`fb_forward_z`
D4	Orthonormality	none (λ=0)	$\mathbb{E}[BB^\top]=I$ (cube λ up to 1000)	`iql_ortho_coef`
D5	Normalization	none	$\|z\|=\|B\|=\sqrt{d}$ sphere / $\sqrt{d}$ scale	`project_z`
D6	Task entry / z-source	raw goal g at train time	$z=\mathbb{E}[rB]$ inferred; $B:=\psi$	`actor_cond`, `actor_z_from_critic`
D7	Actor	flow (FQL) [or DDPG+BC]	flow, greedy on $F^\top z$	`actor_kind`
D8	Task sampling & actor/critic split	value/actor goal *split*: critic mix (traj+cur+rand), actor traj-only	random-direction z; actor & critic *share* one z distribution	`train_goal_ratio`, `decouple_actor_z`

Knob column shows the JAX GCIQL names (`gciql_flow.py`); the FB torch agent (`agents/fb/agent.py`) implements the Direction-B / FB-side rows with parallel knobs under different names — D1 `critic_mode`, D3 `decondition_F`, D5 `d_norm` — and D8's `train_goal_ratio`/`decouple_actor_z` exist *only* there (no `gciql_flow.py` equivalent).

Two differences are bundled inside **D1**: the *backup signal* (reward-value vs. occupancy density) and the *improvement operator* (expectile max vs. none). D1 is the lever; D2–D8 are the structure it rides on.

2 How we built the ladder

Two anchors: **IQL-bilinear** ≈ 90 and **FB-measure** ≈ 34 (@500k; → 48 @1M, still climbing). We morph *one decision at a time* in two directions:

- **Direction A (GCIQL→FB): impose FB structure on the working value.** Start at IQL-bilinear (90) and switch on D2–D5 one at a time. *Does FB's structure survive a value that works?*
- **Direction B (FB→GCIQL): import the value objective into FB.** Start at FB-measure (34) and swap in D1 (and relax D3–D5). *Does the value objective rescue FB's representation?*

The flow actor (D7) is held *fixed* across every state row except the explicit actor isolation (§5). Table 1 is every combination we ran, grouped by direction; §5 then isolates each decision.

Config	D1 obj.	D2 form	FB knobs added (D3–D5)	D7 actor	Succ.
Anchors					
IQL-bilinear $d50$ (base)	IQL	bilinear	–	flow	91.6
GCIQL MLP critic	IQL	MLP	–	flow	71.3
plain FB (measure)	measure	bilinear	z-fwd, ortho, $\sqrt{d}$ (native FB)	flow	34.0‡
Direction A — impose FB structure on IQL-bilinear (D1 held at IQL)					
+ two-tower fwd map	IQL	bil+2-tower	–	flow	86.4
+ two-tower + sphere-Norm	IQL	bil+2-tower	$\sqrt{d}$ sphere (D5)	flow	87.5
+ two-tower + ortho λ=1	IQL	bil+2-tower	ortho_1 (D4)	flow	85.5
+ ortho λ=1000 (pure)	IQL	bilinear	ortho_{1000} (D4)	flow	7.2
B4: all of D2,D4,D5	IQL	bil+2-tower	sphere, ortho_1	flow	**87.1**
Direction B — import the IQL objective into FB / rebuild FB in GCIQL					
FB-IQL, goal-match + GCIQL mix	IQL	bilinear	(FB form kept)	flow	**58.7**
+ de-condition $F{\to}\phi$	IQL	bilinear	drop D3	flow	58.0
+ $\sqrt{d}$ normalization	IQL	bilinear	+D5	flow	65.3
+ both	IQL	bilinear	drop D3, +D5	flow	56.7
+ bilinear-V head	IQL	bilinear	bilinear V	flow	54.0
privileged cube-xyz reward	IQL	bilinear	reward src	flow	62.7
identity-B (z=g) + $\sqrt{d}$	IQL	bilinear	B=I, +D5	flow	65.3
identity-B + de-cond + $\sqrt{d}$	IQL	bilinear	B=I, drop D3, +D5	flow	2.7
z=g on-traj, IQL critic	IQL	bilinear	B=I, on-traj z	flow	50.7
Rebuild real FB inside GCIQL (D1=measure, add structure back)					
measure-only	measure	bilinear	–	flow	9.9
measure-only, ortho_{1000}	measure	bilinear	ortho_{1000}	flow	8.8
measure + z-fwd $F(s,a,z)$	measure	bilinear	D3, ortho_{1000}	flow	25.5
+ sphere-Norm	measure	bilinear	D3, D5, ortho_{1000}	flow	23.1
+ ortho λ=1	measure	bilinear	D3, ortho_1	flow	16.5
C4: z-fwd + sphere + ortho_1	measure	bilinear	D3, D5, ortho_1	flow	**16.5**
Conditioning (D6)					
D1′: actor z from critic ψ	IQL/meas	bilinear	z:=ψ to actor (D6)	flow	22.0

Table 1: **Every combination we ran**, cube-single state, **3 seeds @ 500k**, success (% mean; wandb `amsks/factored-fb`). Every row is seeds $\{0,1,2\}$ read at the 500k eval point. ‡plain FB is 34.0 @500k / 48 @1M — the @500k figure is the one comparable to every other row here. green=near-90, red=collapse.

What the ladder shows at a glance. Direction A stays at 85–88 all the way to B4 — the full FB skeleton survives on the IQL value, the *only* real cost is the two-tower map (90→86) and the one catastrophe is orthonormality at the cube-scale λ=1000 (90→7). Direction B tops out at 59–65

and *rebuilding FB inside GCIQL collapses to* 16–25. Same bilinear form, same flow actor: **D1 (the objective) is the whole gap.**

Cell	one-knob change vs. the scalar-TD anchor	Succ. (%)
Anchors — scalar reward-TD		
IQL bilinear $d50$	the reference cell	91.6 ± 3.1
τ=0.5	drop the expectile improvement operator	88.7 ± 1.0
bilinear $d28$	rank 50→28 (control for identity-ψ)	86.4 ± 7.5
two-tower	FB's two-tower forward map, IQL objective	86.4 ± 0.8
B1 = fb-fwd-z under IQL	task-condition $\phi(s, a, z)$, z=$\psi(g)$	84.4 ± 0.8
Readout crippled — goal leg cannot express the sparse indicator (scalar TD held)		
ψ_{base} frozen	$\psi(g)$ = frozen random MLP	6.1 ± 1.9
ψ_{base} identity	$\psi(g)$=g	2.8 ± 0.9
E3 reward-regressed ψ	ψ trained only by $\langle\psi(s), \psi(g)\rangle \approx r$	0.1 ± 0.3
+ task-conditioned ϕ	E3 + fb-fwd-z	3.5 ± 2.7
SimpleSF grounded (Chua et al.)	ψ trained only by $\langle b, \psi(g)\rangle \approx r$; r *not* readable	0.1 ± 0.3
Objective replaced — successor MEASURE (B×B), λ_{ortho}=1000		
FB measure, ortho_{1000}	fit occupancy density, test states $\sim \rho$	25.5 ± 5.2
+ sphere-Norm	+ $\sqrt{d}$ projection	23.1 ± 3.0
N1: test states $\sim \nu$	measure residual in the *task* norm	20.0 ± 5.3
N2: rank-1 task projection	project residual onto its own z_i=$\psi(g_i)$	6.7 ± 2.2
Objective replaced — successor FEATURE / vector TD		
joint-ψ SF + ortho_{1000}	per-sample vector TD; = FB-measure but diagonal estimator	12.0 ± 1.6
+ absorbing mask (Q=γ^T)	first-hitting instead of visitation density	12.5 ± 1.7
SimpleSF arch (ψ learned)	Chua et al. trunk split, scalar TD, true reward	38.0 ± 6.3
Annealed finetune from the converged 91.6 *geometry* (dose curve, $\lambda(t)$: 0→$\lambda_{\max}$)		
λ=0 control	400k more IQL steps, no SM loss	93.1 ± 7.0
stop-grad-ϕ, λ: 0→10	crosses the cold cliff at λ=3	84.9 ± 2.1
shared legs, λ: 0→100	cold cliff > 10, never located	0.9 ± 0.3
+ fade $\mathcal{L}_{\text{IQL}}$ out	**endpoint = measure-only: does the basin survive?**	5.3 ± 5.2

Table 2: **Follow-up ablations, cube-single state, 3 seeds @ 500k** (mean ±95% CI; wandb `amsks/factored-fb`). Every row is seeds $\{0, 1, 2\}$ at the 500k eval point; runs trained past 500k are read from history at that step. Organized by *what each cell changes*, not by a claimed mechanism. Two failure regimes are separable: a crippled *readout* (goal leg cannot linearly express $\mathbf{1}[s{=}g]$) collapses to ≤ 6 even under the otherwise-perfect scalar TD; a replaced *objective* (successor measure or vector TD) caps at 12–38. The scalar reward-TD is the only configuration that reaches 90. The annealed cells settle whether the cold-start collapse is a property of the objective's fixed point or of the optimization path: warm-starting from the converged 91.6 geometry does *not* rescue the measure. Ramping it in with $\mathcal{L}_{\text{IQL}}$ retained holds 84.9, but fading $\mathcal{L}_{\text{IQL}}$ out — leaving the measure's own fixed point — collapses to 5.3, and the shared-legs ramp to 0.9. The scalar TD term was propping the good basin up; the measure does not have one of its own on cube.

3 Rank is a floor, not a ceiling

A natural reading of the bilinear/MLP gap is that the inner-product head helps by *compressing* — a low-rank bottleneck on the value function. The rank ladder refutes this. Capacity is not being restricted: everything from d=28 to d=512 sits in a flat band, and only d=5 — too few dimensions to span the goal space at all — collapses.

Critic (IQL-trained + flow)	Succ. @500k	
bilinear $d5$	13.3 ± 4.4	rank below the span of the goal set
bilinear $d28$	86.4 ± 7.5	
bilinear $d50$	91.6 ± 3.1	flat band: above the floor,
bilinear $d128$	93.1 ± 0.7	rank is free
bilinear $d512$	87.7 ± 1.4	
MLP critic	71.3 ± 3.4	unconstrained in g; below *every* bilinear rank

At d=512 the goal leg is as wide as the hidden layers, so the bilinear form is barely a capacity restriction at all — and it still beats the MLP by 16 points. What the inner product constrains is not *how much* the critic can express but *how the goal enters*: $Q(s, a, g) = \langle \phi(s, a), \psi(g) \rangle / \sqrt{d}$ with ψ a function of g alone. §4 shows that this is exactly the axis on which the MLP fails.

4 Where the benchmark score is actually lost

The benchmark shifts two things at once relative to the offline data: the episode *start* (the environment's `reset()`, not a dataset state) and the *goal* (an evaluation task, not a state drawn from the data). We re-evaluate every trained agent on the 2×2 that crosses them, on byte-identical start states and goals:

	ID goal (hindsight)	OOD goal (eval task)
ID start (dataset state)	C1	C2
OOD start (`reset`)	C4	**C3** = the benchmark

with Δ_{GOAL}=C2−C1 and Δ_{START}=C4−C1. Measured against the offline data, the *goal* shift is the larger one: eval goals sit 3.6× further from the nearest dataset state than dataset goals do (83.5% beyond the data's p99), against 2.3× (9%) for the start states.

This relocates the failure. The measure objective does not simply make a weaker agent — it makes an agent that is *competent on the data manifold and cannot carry that competence to a new goal.* And it explains the bilinear head: with ψ a function of g alone, $Q(\cdot, \cdot, g)$ for *any* goal lies in the linear span of $Q(\cdot, \cdot, g_i)$ for d dataset goals spanning ψ-space, so a novel evaluation goal needs no extrapolation of the value function — only of the small map ψ. The MLP, nonlinear in g, has no such guarantee, and pays 16 points for it. Consistently, letting the goal leg see the state ($\psi(s, g)$, destroying exactly this property) does not merely revert to MLP quality: it destabilizes the IQL fixed point outright (32.5 ± 39.4; 2/3 seeds diverge late, q crossing positive; the one stable seed lands at 69.2, i.e. MLP level).

5 One decision at a time

Each table below fixes everything except the named decision (bilinear $d50$, flow actor, 3 seeds @ 500k).

D1 — value objective (the lever). Holding form and actor fixed, swap the backup:

Objective on bilinear $d50$ + flow	Succ.
IQL goal-reward value-TD (expectile max)	∼90
measure density, $F(s, a, z)$ (with max-free backup)	25.5
measure density, $\phi(s, a)$	9.9

Swapping IQL→measure on the *same* 90 critic costs ∼ 65–80 points. A genuine z-indexed forward family (D3) lifts the measure floor 9.9→25.5 but gets nowhere near IQL.

	C1 ID/ID	C2 ID/OOD	C4 OOD/ID	C3 **eval**	Δ_{GOAL}	Δ_{START}
Scalar reward-TD, bilinear — the goal shift is free						
bilinear $d512$	93.3	92.7	84.3	96.0	−0.7	−9.0
bilinear $d50$	86.7	93.0	77.3	94.7	+6.3	−9.3
bilinear $d128$	89.0	93.7	82.3	93.3	+4.7	−6.7
Scalar reward-TD, MLP — fits best on-manifold, loses the goal shift						
MLP critic	98.0	82.3	78.7	74.7	−15.7	−19.3
Objective replaced — the goal shift is where it is paid for						
SimpleSF arch	73.0	38.7	65.3	38.3	−34.3	−7.7
measure + z-fwd	45.7	32.7	31.0	25.7	−13.0	−14.7
C4: full FB rebuild	50.0	20.7	26.3	14.7	−29.3	−23.7
joint-ψ SF	67.7	17.7	50.3	13.3	−50.0	−17.3
ortho_{1000} on IQL	64.0	9.7	36.3	5.3	−54.3	−27.7

Table 3: **The 2×2 start $\times$ goal decomposition**, cube-single state, 3 seeds. Success (%). Own re-evaluation harness (H=200, success = $\|\text{cube} - \text{target}\| \leq$ 4cm, the env's own criterion), run on the recorded checkpoints; C3 reproduces the training-time eval for every anchor (MLP 74.7 vs 71.3; bilinear $d50$ 94.7 vs 91.6), which is what licenses reading the other three cells. **Every** bilinear scalar-TD cell in the full table (∼20 of them) has $\Delta_{\text{GOAL}} \in [-0.08, +0.08]$: the goal shift is *free*, and often helps — the five canonical tasks are more structured than random hindsight goals. The MLP loses 16 points to the same shift, and every replaced-objective cell loses 13–54 — while often retaining decent on-manifold competence (joint-ψ SF is 67.7 on C1 and 13.3 on the benchmark). Δ_{START} hurts everyone and separates nobody. *Caveat:* the interaction term is large and positive (+5 to +24) throughout, so the two shifts are not independent; read Δ_{GOAL} as an ordinal signal, not a variance decomposition. FB-family rows are omitted here: their C1/C4 use $z{=}B(g)$ while C2/C3 use reward-inference z, so their deltas would be cross-route and meaningless.

D2 — critic form. IQL objective, exact goal:

Form (IQL-trained + flow)	Succ.
bilinear $\phi^\top \psi / \sqrt{d}$, $d50$	91.6
+ FB two-tower forward map	86.4
MLP critic	71.3

The FB-shaped factorization is *not* the wall (bilinear ≥ MLP); the two-tower map is nearly free.

D4 — orthonormality λ. On IQL-bilinear:

λ (`iql_ortho_coef`)	Succ.
0 (none)	∼90
1 (FB's true coef, w/ 2-tower)	85.5
1000 (cube-scale, pure)	7.2

Orthonormality at λ=1 is harmless; at the cube-scale λ=1000 it *collapses* the IQL critic. FB's representation constraint, at strength, is what a value objective cannot absorb.

D3 / D5 — forward-map z-cond and normalization. Both roughly *flat* (isolations from Table 1): under IQL, sphere-Norm 86.4→87.5 and ortho_1 86.4→85.5 (within noise); under the FB-IQL

import, $\sqrt{d}$ helps a little (58.7→65.3), de-conditioning $F \to \phi$ is flat (58.7→58.0). Structure is second-order; the objective is first-order.

D6 — task entry / z-source. Routing the actor's task vector from the critic's ψ (rather than the value net) under the measure/graft objective scores 22.0. The clean exact-z vs. inferred-z *dissociation* (G4) lives at 1M and is in §9.

D7 — actor. Held fixed across every state row — a control, not a cause. On the *same* bilinear value, flow ≈ 90 vs. DDPG+BC ≈ 55: the flow actor is needed to *realize* the bilinear value, but because it is held identical across every FB-vs-GCIQL comparison it cannot explain the gap.

D8 — goal/task sampling distribution & actor–critic decoupling. The frankenstein knob easiest to leave implicit, so state it plainly. FB and GCIQL disagree not only on the *objective* but on *which tasks the value and the actor ever see*. Canonical FB samples the task z as *random directions* in latent space and shows the *same* distribution to the forward map, the backward map, and the actor: one policy family $\{\pi_z\}$ meant to cover all of latent space. GCIQL never samples a random task — it relabels goals by hindsight from the data and *splits* the distribution: the value learns from a *mix* (trajectory-future goals + the current state, which pins $V{=}0$ at the goal, + some random goals for coverage), while the actor trains *only* on trajectory-future goals, so it practices only *reachable* behaviour. Grafting one agent's objective onto the other's form inherits a free choice here, and it is a real confound: perhaps FB underperforms not because of the measure objective but because its actor is trained to chase random z it can never reach. Two knobs isolate it. **HER task sampling (zHER)** replaces FB's random-direction z with trajectory-future relabelled z, matching FB's task distribution to GCIQL's goals. **Decouple-actor-z** gives the actor GCIQL's trajectory-only distribution while the critic keeps FB's broad one — FB's own version of GCIQL's value/actor split. If the distribution were the lever, either should push FB toward GCIQL. *Neither does:* zHER 32, decouple-actor-z 39 (measure) / 20 (IQL) / 18 (zHER critic goals) — all *below* plain FB (48; all @1M, zHER $n{=}10$ / decouple $n{=}3$). Matching the task distribution to GCIQL, and splitting actor from critic exactly as GCIQL does, never lifts FB above its own ceiling. The distribution is second-order; **D1 is the lever**. (GCIQL's value/actor split is itself already default-ON in every Direction-B row above; these knobs push it *further* toward FB's broad critic distribution and still fail.)

6 Full-stack endpoints

The three endpoints stack their whole cluster; ablations bracket each:

Endpoint	Combination	Ablations	Succ.
B4	IQL + two-tower + sphere + ortho$_1$ (all D2–D5)	87.5 / 85.5	**87.1**
C4	measure + z-fwd + sphere + ortho$_1$ (rebuild FB)	16.5 / 23.1	**16.5**
D1′	actor z from critic ψ (conditioning-source fix)	–	22.0

B4 and C4 are the payoff: **identical architecture, opposite objective** — 87 under IQL, 16 under the measure.[1] No C-ablation escapes the 16–25 band; no B-ablation leaves 85–88.

[1]Scale caveat: the JAX measure cells (`MeasureOnly`/C4/FBz) run at an *uncontrolled* scale — actor q-means of 2.5–4.8k, gradient norms 12–24k, occasional seed crashes — the same unbounded $\langle F, z\rangle$ inflation documented for SG-ISFB (§8). Their numbers are "what the graft does," at the edge of blow-up. A scale-controlled interpolation (measure gradient made dimensionless; grafted onto the working IQL critic at small λ) is underway to separate "the measure gradient destroys value structure" from "it merely fails to build it."

7 What the ladder says

By elimination across the isolations: the **actor** (flow, held fixed, 73 on a good critic), the **bilinear form** (≥MLP), **rank** ($d50$ suffices), **norm** and z**-fwd** (flat) are all exonerated on cube. The binding ingredient is **D1 — the learning objective:** end-to-end IQL value-TD vs. measure-matching, confirmed from both directions (IQL→FB survives to 87; FB→IQL and measure-rebuild cap at 16–65). **Mechanism (proposed):** a measure objective shapes the representation to reconstruct the discounted state-occupancy density, which on manipulation is dominated by where the *arm* goes; the thin, interaction-conditioned successor structure that decides whether a grasp is *kept* through transport is a low-mass correction the density has no incentive to resolve into the value. Value-TD propagates the goal-reaching signal backward through exactly those transitions. **Direct evidence.** The measure's own off-diagonal TD residual is 6347 for the task-blind forward map $\phi(s, a)$ (`MeasureOnly` — it never fits its own objective) versus 266 once the forward map is task-indexed $\phi(s, a, z)$ (`fb_forward_z`), a 24× better density fit — yet that cell still scores only 25.5. The occupancy density is unfittable without a task index, *and fitting it well remains insufficient for control*: precisely the low-mass-correction gap. The fix is not a better basis but an end-to-end value-TD coupling of representation, value, and task — see the `fb-modified` note for objectives that put the max operator back into a successor measure.

Corollary: the IQL critic is zero-shot. If the *objective*, not the representation, is what the measure lacks, the converse should hold — the IQL bilinear critic ought to support FB's own zero-shot API. It does. Recovering a task vector by FB's moment rule from *offline relabelled* transitions, $z = \mathbb{E}_\rho[r(s')\, B(s')]$, and conditioning the flow actor on it recovers 95.5 success — statistically identical to the exact-goal anchor (95.5; 5 tasks × 3 seeds). An earlier "inferred-z → 7" reading was an *on-policy sampling* artefact: a mis-conditioned policy rarely reaches the goal, so its states carry no reward signal to infer from; relabelled *dataset* states fix it. (The actor consumes the raw $B(g)$, so z must be rescaled to $\|B(g)\|$, not $\sqrt{d}$; whitening the non-orthonormal B is refuted — $\mathbb{E}[BB^\top]$ is rank-deficient, $\kappa \approx 10^{14}$.) So the IQL representation already carries FB's task-inference structure: it is the measure *training signal*, not the bilinear form or the inference API, that FB is missing.

8 Putting the max back in the measure: in-sample FB (SG-ISFB)

§1 bundled two things inside D1: the *backup signal* (occupancy density vs. goal-reward) and the *improvement operator* (expectile max vs. none). Every Direction-B row swaps the *whole* of D1 at once. The sharper test is to change *only the operator*: keep FB's successor-measure object, but graft IQL's in-sample max onto it. **If the missing ingredient is just the** max**, this should recover the gap; if it is the reward-TD coupling as a whole, it should not.**

Core insight. Add a *state-only value forward map* $F_V(s, z)$ (architecturally identical to F but fed a zeroed action), giving a value successor measure $N_{ij} = \frac{1}{\sqrt{d}}\langle F_V(s_i, z_i), B(s'_j)\rangle$ alongside the action measure $M_{ij} = \frac{1}{\sqrt{d}}\langle F(s_i, a_i, z_i), B(s'_j)\rangle$. Project both onto the task to get scalar readouts $q_i = \frac{1}{\sqrt{d}}\langle F(s_i, a_i, z_i), z_i\rangle$ (action value) and $v_i = \frac{1}{\sqrt{d}}\langle F_V(s_i, z_i), z_i\rangle$ (state value). The improvement operator is IQL's expectile of v over q, $\mathcal{L}_V = L_2^\tau(\mathrm{sg}\, q - v)$, with advantage $A_i = \mathrm{sg}(q_i - v_i)$. The one load-bearing change is the *bootstrap*: the action measure is fit through the *improved value measure*, $M_{ij} \to \gamma\, \mathrm{sg}\, N^-_{i'j}$ (target net at the next state), with **no** $a' \sim \pi$ — FB's on-policy next-action backup replaced by an in-sample max'd value measure. Zero-shot readout $z_r = \mathbb{E}[r(s)B(s)]$ is untouched, so FB's task inference survives.

Design decisions we tried. Two ways to push the scalar operator into the *vector* measure, plus the incidentals that took several passes to get right:

- **coord** — apply the expectile coordinatewise to every successor column (N as a per-column upper envelope of M). Simplest, but N can be assembled from mutually *incompatible* per-column argmax actions — not the occupancy of any single policy.
- **scalar-gate** — apply the max only to the scalar task value q, then refit the vector N by advantage-weighted regression toward $\mathrm{sg}\, M$ using either **hard** expectile weights $\tau\mathbb{1}[A>0]+(1-\tau)\mathbb{1}[A\leq 0]$ or **AWR** weights $\mathrm{clip}(e^{\beta A}, 0, w_{\max})$. Keeps the operator coherent (one action per row) — the cleaner hybrid on paper.
- **Exact-goal** $z_i{=}B(s_i^{\mathbf{future}})$ (removes the multi-task-dilution confound; the decisive incidental), **goal-success masking** of the measure-TD bootstrap (GCIQL-consistent), **ortho** $\lambda{=}1$ (FB-scale, not cube-1000), and **std-normalized AWR advantage**: the FB scalar $\langle F, z\rangle$ *inflates without bound* over training (q, A_{std} into the thousands as F grows and B stays norm-capped), so a fixed β re-saturates the $w_{\max}$ clip and AWR collapses; normalizing makes β scale-invariant.

Result (cube-single state, 3 seeds @ 500k, torch).

Variant	Operator into the measure	Succ.
C4 (failed baseline)	standard FB measure, no operator	16.5
plain FB (reference)	native measure	34 (48 @1M)
coord IS-FB	coordinatewise expectile envelope	**33.3**
SG-ISFB, AWR weights	scalar gate, smooth $e^{\beta A}$	26.7
SG-ISFB, hard weights	scalar gate, hard expectile	8.0
B4 (target)	IQL backup on FB skeleton	87.1

What it says. The operator *helps the measure but nowhere near enough.* `coord` roughly *doubles* the C4 floor (16.5→33) — inserting an in-sample max into the rebuilt-FB measure is a real lift — but it lands right at plain FB's own 500k level (34), below even the FB-IQL *import* band (59–65), and less than half of B4 (87). Two further surprises. **(i)** The scalar-gated refinement, argued to be the cleaner hybrid, *underperforms* the coordinatewise baseline it was meant to replace (33→27 AWR / 8 hard); on this substrate the incoherent-but-optimistic per-column envelope beats the coherent scalar gate, and `hard` weights collapse. **(ii)** The measure scale inflates unbounded (F grows, B norm-capped) — benign for the scale-invariant `coord`/`hard` operator and the FQL actor, but it is what forced the AWR std-norm fix and is the standing suspect for the `hard` collapse.
In the terms of §7: SG-ISFB is the "*improves but plateaus below B4*" outcome. The max operator is *not* the whole of D1 — putting it back into the successor measure lifts FB off the floor but does not recover reward-TD's value-controllability. The binding ingredient remains the end-to-end *reward-TD coupling* of representation, value, and task, not the improvement operator in isolation. This *sharpens* §7 rather than softening it: even the one modification §7 flagged as the promising fix leaves FB in its own band.

9 [BEING UPDATED] 1M / 10-seed confirmation tier

This section is the slow, confirmed tier (1M steps, n=8–10 seeds). It is being refreshed at the corrected 500k incidentals; where it disagrees with the main body, the 500k ladder above is the current best evidence. Numbers retained from the 2026-06-22 10-seed×1M campaign.

Grid (1M, 10 seeds). Base bilinear $d50$ **92.6** $\pm$ **1.3**; τ=0.5 88.9, τ=0.95 90.6, single-Q 90.2 (pessimism knobs flat); rank $d5$ 14.3, $d50$ 92.6, $d128$ 93.9, $d512$ 94.5 (capacity cliff far below FB's operating

rank); DDPG+BC actor 56.6. FB side: plain FB 48, zHER 32, FB-IQL critic (learned B) 23.6, +goal-TD q-loss 38.8. CRL 25.8 (flow) / 16.8 (Gaussian).

G4 — success-level objective isolation (a dissociation, n=9).

representation objective	actor conditioning	Succ.
measure (plain FB)	zero-shot z	48 ± 5
measure (FB-IQL critic)	zero-shot z	24 ± 7
IQL (G4)	**exact** $z{=}\psi(g)$	$\mathbf{93 \pm 2}$
IQL (G4)	**zero-shot** $z{=}\mathbb{E}[r\psi]$	$\mathbf{7 \pm 3}$
IQL (bridge)	exact goal (raw)	94 ± 1

Two separable layers: (i) the **objective** sets whether the value is controllable at all (the 48→93 jump); (ii) the **inference→actor** path is an independent success-level bottleneck once zero-shot (7 despite cf-AUC 0.92). The clean next control is an actor trained across the inferred-z distribution. Value-level probes (cf-AUC 0.92 exact ≈ inferred; "improving the representation" via IWR/PSM/d128 *lowers* success while raising interaction lift-R^2) are retained in the verification log and fold into the same conclusion: inside the FB family there is no value-consumer for interaction information.